

Statement of Requirements (SOR)

For

Integrated Electro-Optical/Infrared (EO/IR) Target Projector System

1.0 REQUIREMENT OVERVIEW

The Air Force Test Center (AFTC) has a mission-critical requirement for a new, turnkey Electro-Optical/Infrared (EO/IR) Target Projector System. This system is essential for the laboratory characterization, performance verification, and calibration of advanced sensor systems. The Government requires a single, fully integrated system from one vendor to eliminate technical risk associated with integrating disparate components and to ensure data integrity through a unified software control platform.

2.0 SCOPE

The scope of this requirement is for the delivery of a complete hardware and software system, including on-site installation, calibration, and operator training. The system must be capable of simulating a wide range of infrared signatures with exceptionally high fidelity and radiometric accuracy.

3.0 SPECIFIC REQUIREMENTS

The system delivered shall meet or exceed the following minimum Government requirements:

Requirement ID	Category	Minimum Requirement
SR-1	System Performance	The system shall be capable of projecting targets across the visible spectrum through the Long-Wave Infrared (LWIR) spectrum (approximately 0.4μm to 14μm).
SR-2	Radiometric Accuracy	The system must include at least two blackbody infrared sources, each demonstrating a minimum emissivity of > 0.995. One source shall be a differential type, and the other shall be an absolute type.
SR-3	Visible/SWIR Capability	The system must include an integrated uniform source (e.g., an integrating sphere) for sensor testing in the visible through Short-Wave Infrared (SWIR) spectrums.
SR-4	Hardware Integration	The system must be delivered as a complete, factory-integrated turnkey solution, including all sources, collimator, motorized target wheels, and system controller rack. The Government will not act as a systems integrator.

SR-5	Software Integration	All hardware components must be controllable from a single, unified software interface provided with the system. The software shall be delivered fully installed, configured, and capable of executing automated test sequences.
SR-6	Support Services	The vendor shall provide on-site installation, final system alignment, and complete functional verification at the Government facility.
SR-7	Training	The vendor shall provide on-site operator training for up to eight (8) Government personnel, covering system operation, maintenance, and use of the control software.
SR-8	Warranty	The entire system shall be covered by a standard commercial warranty of no less than 12 months, effective upon Government acceptance.

4.0 DELIVERABLES

The contractor shall deliver the following:

Item	Description
1	One (1) complete, integrated EO/IR Target Projector System meeting all requirements in Section 3.0.
2	One (1) perpetual software license for the system control software.
3	On-site installation and calibration services.
4	On-site operator training course.
5	Complete set of commercial operator and maintenance manuals.
6	A Certificate of Conformance certifying the system meets all stated requirements.

5.0 PLACE OF PERFORMANCE

System fabrication will occur at the contractor's facility. Final installation, training, and acceptance will occur at: Edwards AirForce Base, CA.

6.0 GOVERNMENT-FURNISHED PROPERTY

The Government will provide one (1) optical table for system installation. The contractor shall provide detailed specifications for this table within 60 days of contract award.